Problem Statement.

ROBOT BASE ARCH LIKE CROWD CONGETION REDUCTION AT NARROW EXITS.

According to paper[1]'s solution it try to reduce the bottleneck. where a "arch" like structure generate around the exit. they suggest that instead of forming the arch at exit point, the arch would be born the little upstream. by establishing two pillars. making Arch wide & fragile.

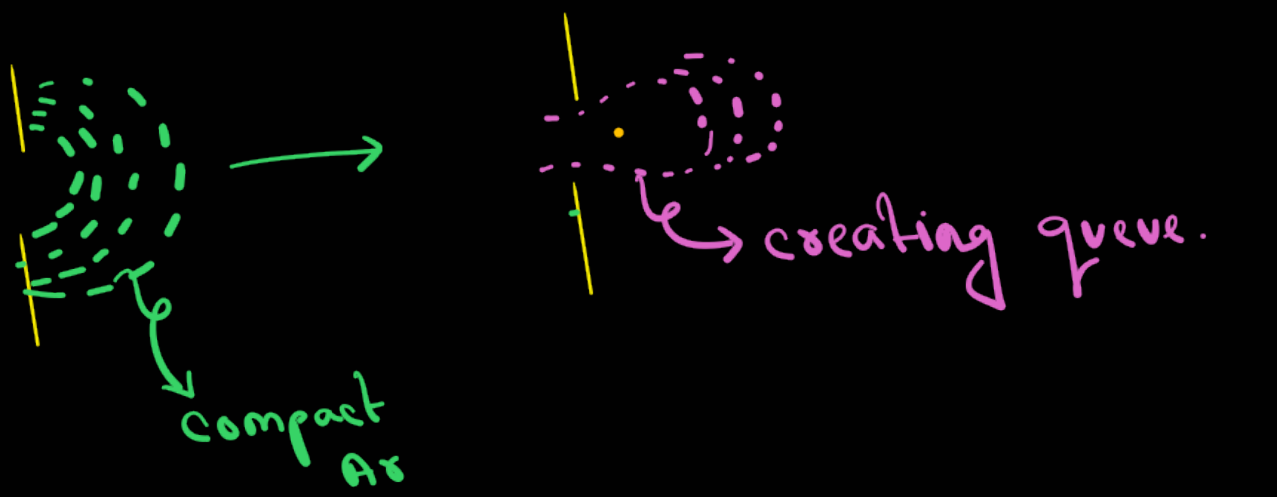


so,
we proposed this problem in a dynamic way.

1. Replace pillars with robots.
2. The architecture of our solution is not make arch fraghile but Never it to form
3. By converting the crowd in queue or lanes. of pedestrian.

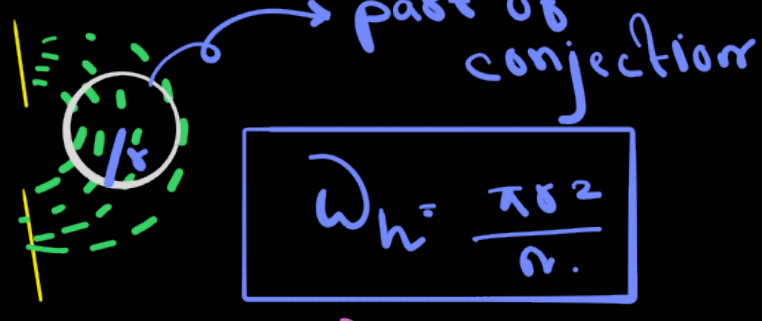
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4. The placing of robot will at exit but it will depends on threshold.



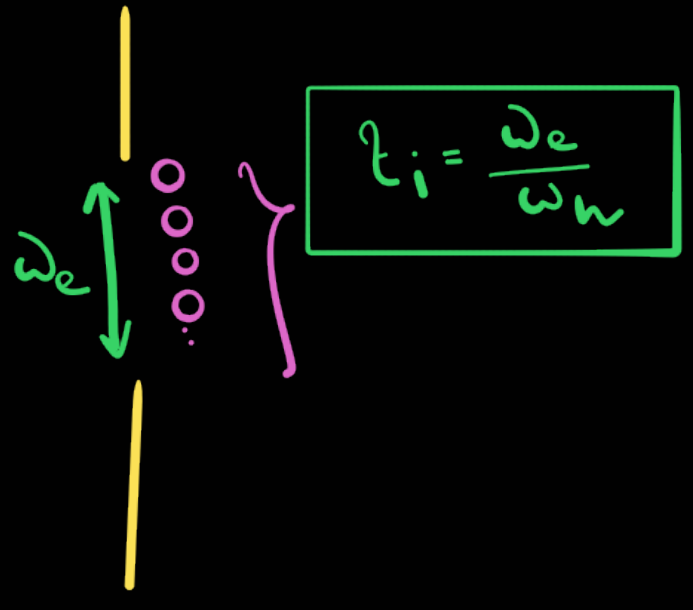
FORMULATION:→

* Threshold (t_A)
* Avg-human width. (ω_h)



$$\omega_h = \frac{\pi r^2}{n}$$

Gdial threshold (t_i)



$$t_i = \frac{\omega_e}{\omega_h}$$

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→ Replace with robot

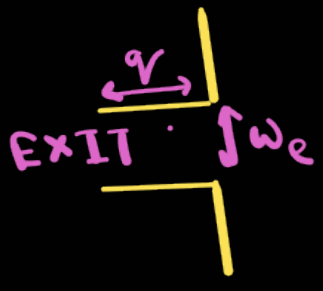
→ Robot

Assuming Robot (ω_r) is same as ω_h

$$\omega_r = \omega_h$$

$$\eta_A = \lceil t_i/2 \rceil = \left\lceil \frac{\omega_e}{2\omega_h} \right\rceil$$

⊛ Throughput (T)



Let say a pedestrian can cover exit in p unit time.

Ⓗ q unit covers in p unit time.

$$T = \lceil \eta_A / p \rceil \text{ pedestrian/unit time.}$$

⊛ what is a voronoi diagram?

A voronoi diagram divids an area into regions based on the positions of individual points.

⊛ Every region belongs to one pedestrian.

⊛ Every location inside a region is closer to its owner than to any other pedestrian.

⊛ The boundaries are formed where locations are equally distant from neighboring.

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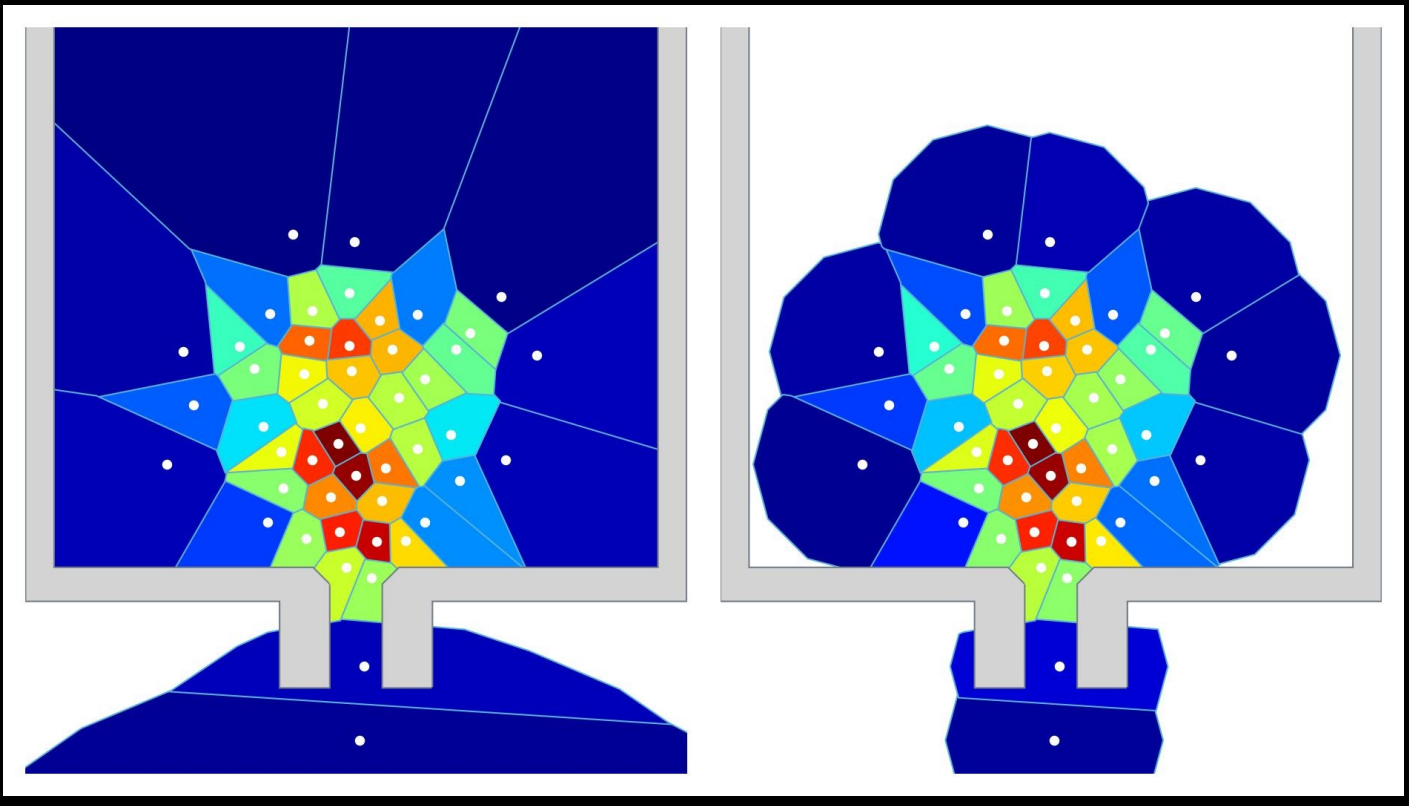
Each region is called a Voronoi Cell.

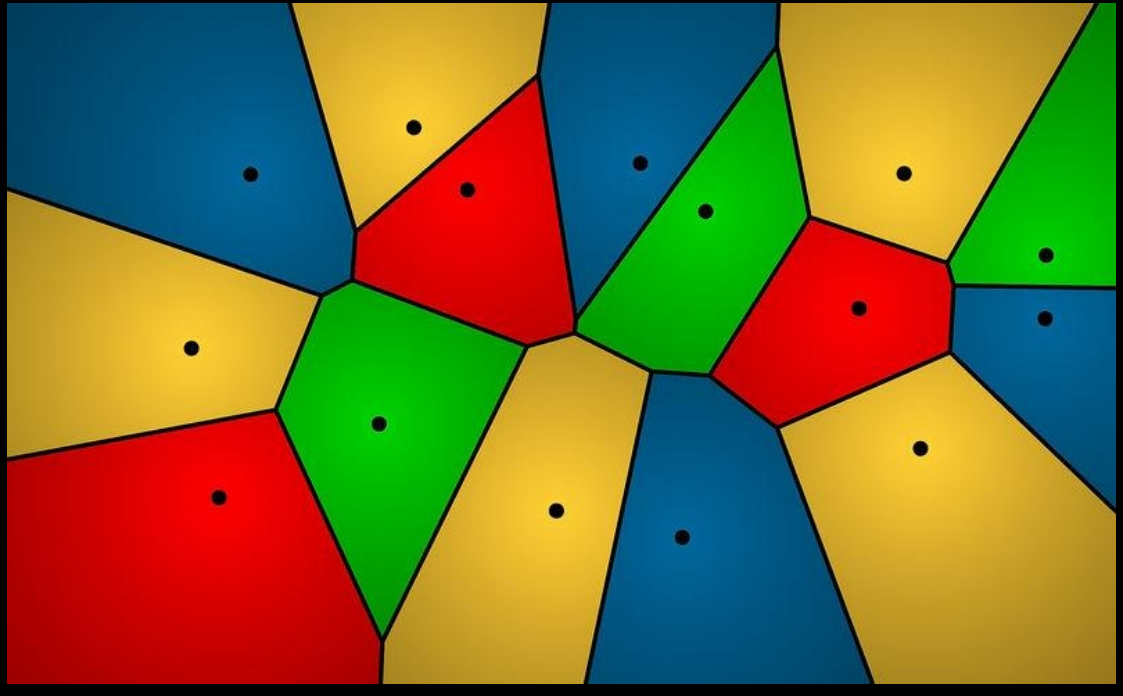
$$D_{V'} = \frac{N}{\sum_{i=1}^{N} |A_i|}$$

$$A_i' = A_i \cap W$$

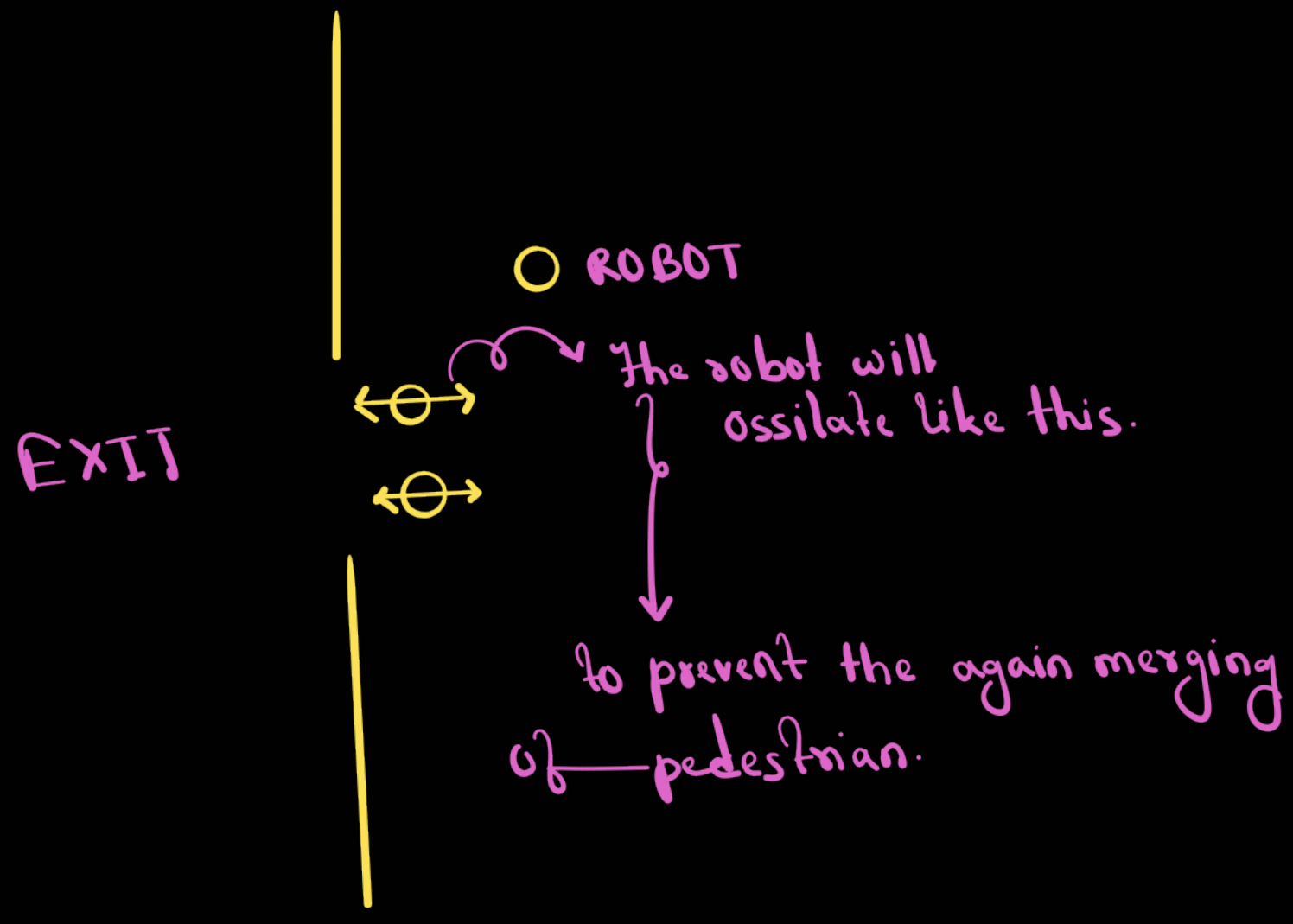
N: number of pedestrians

A_i: Voronoi cell belonging to pedestrian i.





ROBOT ARCHITECTURE.



① The robot
- ✱ has it own brain by ORL (Deep Reinforcement learning)
- ✱ And Adapts it'self by ADP (Adaptive dynamic programming) - in [paper 1]

✱ There will be a central control System & overheaded camera.

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REFERANCE :

[1] Arch-Research paper.pdf

[2] ADP-Robot paper.pdf

[3] density-calculation paper.pdf.

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